

Assignment for Week 5

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Assignment

1) 14.1.2 (d) (p. 274)

Find the gradient vector and the Hessian matrix of the function $f(x, y)$, and evaluate them at $(1, -2)$.

Answer: For $f(x, y)$, we have the gradient vector

$$\begin{aligned}\nabla f(x, y) &= (f_x, f_y) \\ &= (\ln(1 + y^2), \frac{2xy}{1 + y^2}),\end{aligned}$$

and the Hessian matrix

$$\begin{aligned}D^2 f(x, y) &= \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} \\ &= \begin{bmatrix} 0 & \frac{2y}{1+y^2} \\ \frac{2y}{1+y^2} & \frac{2x(1-y^2)}{(1+y^2)^2} \end{bmatrix}.\end{aligned}$$

Therefore, we have

$$\nabla f(1, -2) = (\ln 5, -\frac{4}{5}) = (\ln 5, -0.8),$$

and

$$D^2 f(1, -2) = \begin{bmatrix} 0 & -\frac{4}{5} \\ -\frac{4}{5} & -\frac{6}{25} \end{bmatrix} = \begin{bmatrix} 0 & -0.8 \\ -0.8 & -0.24 \end{bmatrix}.$$

2) 14.2.2 (p. 279)

If

$$z = x^3 y^4, \quad x = 1 + 6t, \quad y = -2 - 3t,$$

use the chain rule to find dz/dt .

Answer: By the chain rule, we have

$$\begin{aligned} \frac{dz(x(t), y(t))}{dt} &= \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \\ &= (3x^2 y^4 + e^y) \cdot 6 + (4x^3 y^3 + x e^y) \cdot (-3) \\ &= 6(3x^2 y^4 + e^y) - 3(4x^3 y^3 + x e^y). \end{aligned}$$

We can express the result in terms of t :

$$\begin{aligned} \frac{dz(x(t), y(t))}{dt} &= 6(3x^2 y^4 + e^y) - 3(4x^3 y^3 + x e^y) \\ &= 6x^2 y^3 (3y - 2x) - 3e^y (2 - 3x) \\ &= 6(6t + 1)^2 (3t + 2)^3 (21t + 8) + \frac{3(18t + 1)}{e^{3t+2}}. \end{aligned}$$

3) 16.1.5 (a) (p. 323)

Find the critical points of the function $z = 3x^2 - y^3 + 12xy - 36y$ and classify each as a local maximum, a local minimum, a saddle point or none of these.

Answer: For $z = 3x^2 - y^3 + 12xy - 36y$, we have the gradient vector

$$\nabla z(x, y) = (6x + 12y, -3y^2 + 12x - 36),$$

and the Hessian matrix

$$D^2 z(x, y) = \begin{bmatrix} 6 & 12 \\ 12 & -6y \end{bmatrix}.$$

Let $\nabla z(x, y) = 0$, we have $(x, y) = (4, -2)$ or $(x, y) = (12, -6)$. Hence the critical points of $z(x, y)$ are $(4, -2, 32)$ or $(12, -6, 0)$.

At $(x, y) = (4, -2)$, the Hessian matrix is $D^2 z(4, -2) = \begin{bmatrix} 6 & 12 \\ 12 & 12 \end{bmatrix}$. Since $6 > 0$, $12 > 0$, $\det(D^2 z(4, -2)) = -72 < 0$, the Hessian matrix is neither positive or negative (semi)definite. Therefore, $(4, -2, 32)$ is a saddle point of $z(x, y)$.

At $(x, y) = (12, -6)$, the Hessian matrix is $D^2 z(12, -6) = \begin{bmatrix} 6 & 12 \\ 12 & 36 \end{bmatrix}$. Since $6 > 0$, $36 > 0$, $\det(D^2 z(12, -6)) = 0$, the Hessian matrix is positive semidefinite. Therefore, $(12, -6, 0)$ is a local minimum of $z(x, y)$.

4) 16.2.6 (p. 331)

Show that the function $f(x, y, z) = e^x + e^y + e^z$ is convex.

Answer: We first show that $g(x) = e^x$ is convex. Then it follows that $f(x, y, z) = e^x + e^y + e^z$ as a sum of convex functions is also convex.

The Hessian matrix of $g(x) = e^x$ is $D^2g = \begin{bmatrix} e^x & e^x \\ e^x & e^x \end{bmatrix}$. Since $e^x > 0$ for all $x \in \mathbb{R}$, and the determinant of D^2g is 0 for all $x \in \mathbb{R}$, D^2g is positive semidefinite for all $x \in \mathbb{R}$.

Therefore, $g(x) = e^x$ is convex, and it follows that $f(x, y, z) = e^x + e^y + e^z$ is convex.

5) 17.2.1 (p. 349)

Find the maximum and minimum values of $2x + 3y + z$ subject to

$$x^2 + y^2 + z^2 = 9.$$

Answer: Let $f(x, y, z) = 2x + 3y + z$ and $g(x, y, z) = x^2 + y^2 + z^2 - 9$. Define the Lagrangian of $f(x, y, z)$ as

$$L(x, y, z, \lambda) = f(x, y, z) - \lambda g(x, y, z),$$

where $\lambda \in \mathbb{R}$ is a constant. Then the gradient vector of $L(x, y, z, \lambda)$ is

$$\nabla L(x, y, z, \lambda) = (2 - 2\lambda x, 3 - 2\lambda y, 1 - 2\lambda z, -x^2 - y^2 - z^2 + 9).$$

Let $\nabla L(x, y, z, \lambda) = 0$, we have $x = \frac{1}{\lambda}$, $y = \frac{3}{2\lambda}$, and $z = \frac{1}{2\lambda}$. Hence $f(x, y, z) = \frac{7}{\lambda}$.

By the constraint $x^2 + y^2 + z^2 = 9$, we have $\lambda = \pm \frac{\sqrt{14}}{6}$. Therefore, the maximum and minimum of $2x + 3y + z$ are $3\sqrt{14}$ and $-3\sqrt{14}$.

6) Using `mapply()`, the class of `PersonnelData`'s columns.

```
PersonnelData <- data.frame(Representative = c(1:4),
                             Sales = c(95,110,115,90), Territory = c(1:4))
mapply(class,PersonnelData)
```

```
## Representative      Sales      Territory
##      "integer"      "numeric"      "integer"
```

We know that the class of `PersonnelData`'s columns are, respectively, integer, numeric, and integer.

7) Using the ideas from the Foundations of Regression sheet, why are the following two regressions different?

Note, `-1` means leave out the intercept. Regressing on `1` would recover just the intercept.

Reason: `lm( y ~ x - 1 )` is the regression of y onto x , while `lm( x ~ y - 1 )` is the regression of x onto y .

```
x <- 1:10
y <- 1/3*x + rnorm(10)
lm( y ~ x - 1 )
```

```
##
## Call:
## lm(formula = y ~ x - 1)
##
## Coefficients:
##          x
## 0.3234
```

```
lm( x ~ y - 1 )
```

```
##
## Call:
## lm(formula = x ~ y - 1)
##
## Coefficients:
##          y
## 2.891
```

8) Use `%*%` to get the coefficients from the regression from the previous problem.

From the Foundations of Regression sheet, the coefficient in the bivariate case is

$$\beta = \frac{Cov(x, y)}{Var(x, y)},$$

and that $Cov(x, y) = E(x * y) - E(x)E(y)$, and $Var(x) = E(x * x) - E^2(x)$. Therefore, we have the following:

```
# To get the coefficients from lm(y~x-1)
(mean(x%*%y)-mean(x)*mean(y))/(mean(x%*%x)-(mean(x))^2)
```

```
## [1] 0.3228596
```

```
# To get the coefficients from lm(x~y-1)
(mean(y%*%x)-mean(y)*mean(x))/(mean(y%*%y)-(mean(y))^2)
```

```
## [1] 2.879471
```

The results are close to the ones in Problem 7).

- 9) Install the fueleconomy package. Using the data(vehicles) dataset, use lm to recover the average number of cylinders.

```
library(fueleconomy)
str(vehicles)
```

```
## Classes 'tbl_df', 'tbl' and 'data.frame':  33442 obs. of  12 variables:
## $ id   : int  27550 28426 27549 28425 1032 1033 3347 13309 13310 13311 ...
## $ make : chr   "AM General" "AM General" "AM General" "AM General" ...
## $ model: chr   "DJ Po Vehicle 2WD" "DJ Po Vehicle 2WD" "FJ8c Post Office" "FJ8c Post Office" ...
## $ year : int   1984 1984 1984 1984 1985 1985 1987 1997 1997 1997 ...
## $ class: chr   "Special Purpose Vehicle 2WD" "Special Purpose Vehicle 2WD" "Special Purpose Vehicle :
## $ trans: chr   "Automatic 3-spd" "Automatic 3-spd" "Automatic 3-spd" "Automatic 3-spd" ...
## $ drive: chr   "2-Wheel Drive" "2-Wheel Drive" "2-Wheel Drive" "2-Wheel Drive" ...
## $ cyl   : int    4 4 6 6 4 6 6 4 4 6 ...
## $ displ: num    2.5 2.5 4.2 4.2 2.5 4.2 3.8 2.2 2.2 3 ...
## $ fuel  : chr   "Regular" "Regular" "Regular" "Regular" ...
## $ hwy   : int   17 17 13 13 17 13 21 26 28 26 ...
## $ cty   : int   18 18 13 13 16 13 14 20 22 18 ...
```

```
# The column name of "cylinder" is "cyl"
# In place of the formula of lm(), we put vehicles$cyl~ 1, and the regression of
# all the values of cylinders will be the average number of cylinders we want to find.
lm(vehicles$cyl~ 1)
```

```
##
## Call:
## lm(formula = vehicles$cyl ~ 1)
##
## Coefficients:
## (Intercept)
##          5.772
```

```
# The average number of cylinders is hence 5.772.
# To test our result:
mean(vehicles$cyl, na.rm = TRUE)
```

```
## [1] 5.771867
```

- 10) Use `lm` to recover the average number of cylinders by fuel type. Check your answer with `dplyr` `group_by` and `summarise`.

```
# The column names of "cylinder" and "fuel type" are respectively "cyl" and "fuel".  
# In place of the formula of lm(), we put vehicles$cyl~vehicles$fuel-1, and the regression  
# of cylinders onto fuel will produce the average numbers of cylinders by fuel type that we want to find  
lm(vehicles$cyl~vehicles$fuel-1)
```

```
##  
## Call:  
## lm(formula = vehicles$cyl ~ vehicles$fuel - 1)  
##  
## Coefficients:  
##                vehicles$fuelCNG  
##                6.517  
##                vehicles$fuelDiesel  
##                6.365  
##                vehicles$fuelGasoline or E85  
##                7.083  
##                vehicles$fuelGasoline or natural gas  
##                6.889  
##                vehicles$fuelGasoline or propane  
##                8.000  
##                vehicles$fuelMidgrade  
##                7.953  
##                vehicles$fuelPremium  
##                6.363  
##                vehicles$fuelPremium and Electricity  
##                6.000  
##                vehicles$fuelPremium Gas or Electricity  
##                4.571  
##                vehicles$fuelPremium or E85  
##                7.500  
##                vehicles$fuelRegular  
##                5.450  
##                vehicles$fuelRegular Gas and Electricity  
##                4.000
```

```
# To test our result by group_by and summarise:  
vehicles %>% na.omit() %>% group_by(fuel) %>% summarise(meancyl=mean(cyl))
```

```
## # A tibble: 12 × 2  
##               fuel meancyl  
##               <chr>   <dbl>  
## 1                CNG 6.517241  
## 2                Diesel 6.364989  
## 3      Gasoline or E85 7.083413  
## 4 Gasoline or natural gas 6.888889  
## 5      Gasoline or propane 8.000000  
## 6                Midgrade 7.953488  
## 7                Premium 6.362771  
## 8      Premium and Electricity 6.000000  
## 9      Premium Gas or Electricity 4.571429  
## 10               Premium or E85 7.500000  
## 11                Regular 5.449887  
## 12 Regular Gas and Electricity 4.000000
```

- 11) Using the vehicles dataset, estimate the relationship (slope coefficient) between the year of introduction and highway fuel economy for each class of vehicle, and plot the coefficients using `geom_points`, add `geom_errorbar` based on 2 times the standard error estimates of the coefficients.

```
# The column representing highway fuel economy is "hwy".
# We first split the vehicles dataset by class, and create a list of 34 elements.
splitclass <- vehicles %>% split(.$class)
# We then summarise the regression coefficients of "hwy" on to "year" by different classes:
sumveh <- cbind(as.data.frame(
  splitclass %>%
  map( ~ lm(hwy ~ year, data=.x)) %>%
  map(summary) %>%
  map(c("coefficients"))%>%
  map_dbl(2)),
  as.data.frame(
  splitclass %>%
  map( ~ lm(hwy ~ year, data=.x)) %>%
  map(summary) %>%
  map(c("coefficients"))%>%
  map_dbl(2)))
```

```
## Warning in summary.lm(.x[[i]], ...): essentially perfect fit: summary may
## be unreliable
```

```
## Warning in summary.lm(.x[[i]], ...): essentially perfect fit: summary may
## be unreliable
```

```
names(sumveh) <- c("Coefficient", "Standard Error")
sumveh$numclass <- seq(1:length(sumveh$Coefficient))
# ggplot() + geom_point(data=sumveh, mapping=aes(x=sumveh$numclass, y=sumveh$coefficients))
sumveh
```

##	Coefficient	Standard Error	numclass
## Compact Cars	0.151158104	0.151158104	1
## Large Cars	0.132901251	0.132901251	2
## Midsize-Large Station Wagons	0.086250077	0.086250077	3
## Midsize Cars	0.262026644	0.262026644	4
## Midsize Station Wagons	0.173644493	0.173644493	5
## Minicompact Cars	0.113032210	0.113032210	6
## Minivan - 2WD	0.252802193	0.252802193	7
## Minivan - 4WD	0.081568627	0.081568627	8
## Small Pickup Trucks	-0.151066369	-0.151066369	9
## Small Pickup Trucks 2WD	-0.081514586	-0.081514586	10
## Small Pickup Trucks 4WD	0.001059736	0.001059736	11
## Small Sport Utility Vehicle 2WD	0.260973370	0.260973370	12
## Small Sport Utility Vehicle 4WD	0.696107625	0.696107625	13
## Small Station Wagons	0.059028966	0.059028966	14
## Special Purpose Vehicle	NaN	NaN	15
## Special Purpose Vehicle 2WD	0.206751383	0.206751383	16
## Special Purpose Vehicle 4WD	0.018133512	0.018133512	17
## Special Purpose Vehicles	0.069120645	0.069120645	18
## Special Purpose Vehicles/2wd	1.500000000	1.500000000	19
## Special Purpose Vehicles/4wd	0.000000000	0.000000000	20
## Sport Utility Vehicle - 2WD	0.410638506	0.410638506	21
## Sport Utility Vehicle - 4WD	0.336283192	0.336283192	22

## Standard Pickup Trucks	0.117774862	0.117774862	23
## Standard Pickup Trucks 2WD	0.076203360	0.076203360	24
## Standard Pickup Trucks 4WD	0.076946861	0.076946861	25
## Standard Pickup Trucks/2wd	0.866025404	0.866025404	26
## Standard Sport Utility Vehicle 2WD	0.684521465	0.684521465	27
## Standard Sport Utility Vehicle 4WD	0.607664234	0.607664234	28
## Subcompact Cars	0.031654992	0.031654992	29
## Two Seaters	0.089393536	0.089393536	30
## Vans	-0.047449218	-0.047449218	31
## Vans Passenger	-1.200000000	-1.200000000	32
## Vans, Cargo Type	0.015720943	0.015720943	33
## Vans, Passenger Type	0.015085329	0.015085329	34

I encounter some problem when trying to plot the regression by class.